

■ CSS Animations - Learning Notes

@keyframes

Defines the steps of animation. You can use `from` → `to` or percentage (`0%`, `50%`, `100%`).

```
@keyframes changecolor {
  0%   { background-color: red; }
  25%  { width: 300px; }
  50%  { background-color: orange; }
  75%  { width: 200px; }
  100% { background-color: yellow; }
}
```

Applying Animation

Use `animation-name`, `animation-duration`, `animation-iteration-count`, etc.

```
#animate {
  background-color: red;
  height: 200px;
  width: 200px;
  animation-name: changecolor;
  animation-duration: 4s;
  animation-iteration-count: infinite;
}
```

Animation Direction & Timing

Control the direction and speed curve of animation.

```
#animate2 {
  animation-name: changecolor;
  animation-duration: 4s;
  animation-iteration-count: infinite;
  animation-timing-function: ease;
  /* Options: ease, linear, ease-in, ease-out, ease-in-out */
}
```

Complex Animation with Shape

Multiple changes: background-color + border-radius.

```
@keyframes chnges {
  0%   { background-color: red; border-radius: 10%; }
  50%  { background-color: green; border-radius: 50%; }
  100% { background-color: cyan; border-radius: 0%; }
}

#animate3 {
  height: 300px;
  width: 300px;
  animation-name: chnges;
  animation-duration: 2s;
  animation-iteration-count: infinite;
  animation-timing-function: linear;
}
```

■ Summary

- @keyframes = steps of animation

- animation-name = connect keyframes to element
- animation-duration = time for one cycle
- animation-delay = wait before start
- animation-iteration-count = repeat count or infinite
- animation-direction = normal, reverse, alternate
- animation-timing-function = ease, linear, ease-in, ease-out
- Animations can change color, size, shape, border-radius, etc.